



EE 517: Wireless Communications and Networking

Course Code: EE 517

Course Name: Wireless Communications and Networking

L-T-P-C: 3-0-0-3

Prerequisites: EE-304 Communication Theory, EE-503 Advance Communication Theory or their equivalents, or the instructors consent

Intended for: UG/PG

Distribution: Elective

Approval: 9th Senate

Course Contents

- **Review of Information Theory, and Detection and Estimation Theory:** channel coding theorems, capacity of various channel models; detection and estimation in additive Gaussian noise. [3 Lectures]
- **Wireless Channel:** Path loss, Physical modeling, I/O models, Channel models. [4 Lectures]
- **Point to Point Communication:** Detection in Rayleigh fading channels, Diversity: time, frequency, antenna, receiver and transmitter; Impact of channel uncertainty. [5 Lectures]
- **Capacity of Wireless Channels:** AWGN channel capacity, LTI Gaussian channels, Capacity of fading channels. [4 Lectures]
- **Multiuser Channel Capacity:** Up/down-link AWGN channel, Up/down-link fading channel, frequency selective fading channels, multiuser diversity. [5 Lectures]
- **MIMO Channels:** Physical modeling, Slow and fast fading channels, Capacity, Multiplexing architectures: V-BLAST and D-BLAST, Diversity gain, Diversity-multiplexing tradeoff and universal code design for optimal DM tradeoff, Multiuser communication. [9 Lectures]
- **Cellular Systems:** Channel reuse, Multiple access, Interference management, Dynamic resource allocation: Cognitive Radio, Narrowband systems: GSM, Wideband systems: OFDM and CDMA, Cellular phone standards. [9 Lectures]
- **Other Topics:** Wireless LANs: Aloha, CSMA/CA, Protocols, Mesh networks: capacity, routing, and scheduling. [3 Lectures]

Reference Books:

1. D. Tse and P. Viswanath, **Fundamentals of Wireless Communication**, Cambridge Univ. Press, 2005.
2. A. Goldsmith, **Wireless Communications**, Cambridge Univ. Press, 2005.
3. A. Kumar, D. Manjunath, and J. Kuri, **Wireless Networking**, Morgan Kaufmann, 2008.
4. B. Clerckx and C. Oestges, **MIMO Wireless Networks**, Academic Press, 2/e, 2013.
5. Related research papers.